

Finite Sample Properties of FGLS and White's Heteroskedasticity Procedures: A Monte Carlo Study

Simulation based evidence on estimation, testing and robust variance
estimation

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2026

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1 Introduction

The finite-sample performance of econometric estimators and hypothesis tests may differ from their asymptotic properties, particularly when the assumptions underlying standard estimation and inference procedures are not satisfied. Monte Carlo simulations provide a useful framework for studying these properties by allowing estimators and test statistics to be evaluated repeatedly under controlled data-generating processes.

This study examines the finite-sample properties of estimation and inference procedures in the presence of heteroskedasticity. The analysis is organized around two main components. First, the behavior of the Feasible Generalized Least Squares (FGLS) estimator is evaluated under a known heteroskedastic covariance structure. The analysis considers the sampling behavior of the coefficient estimators, as well as the empirical size and power of hypothesis tests across different sample sizes. In addition, the equivalence between FGLS and Ordinary Least Squares (OLS) applied to the model transformed through a Cholesky decomposition is examined.

Second, the finite-sample performance of White's test for heteroskedasticity is analyzed under different combinations of error distributions and variance structures. The empirical size of the test is evaluated under homoskedastic and normally distributed errors, while its power is studied under heteroskedasticity with both normal and non-normal disturbances. The analysis also evaluates White's heteroskedasticity-robust covariance estimator by comparing its estimated variances with the corresponding Monte Carlo variances of the OLS coefficient estimators.

The simulations are conducted for several sample sizes, with 5,000 replications for each experimental configuration. This allows the analysis to characterize how the performance of the estimators and tests changes as the sample size increases, and to assess the effects of heteroskedasticity and non-normality on finite-sample inference.

Overall, the objective is to provide a systematic computational evaluation of the finite-sample behavior of FGLS, White's heteroskedasticity test, and White's robust variance estimator under the data-generating processes considered in the study.

2 Theoretical Framework

2.1 Linear Regression and Non-Spherical Errors

Consider the linear regression model

$$y = X\beta + u, \tag{1}$$

where y is the $n \times 1$ vector of observations, X is the $n \times K$ matrix of explanatory variables, β is the vector of parameters, and u is the error vector.

Under the standard exogeneity condition,

$$E(u \mid X) = 0, \tag{2}$$

the OLS estimator is

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'y. \tag{3}$$

When the errors are spherical, the conventional variance-covariance matrix of the OLS estimator is valid. However, in the presence of heteroskedasticity, the error covariance matrix is no longer proportional to the identity matrix. In this case, the conventional OLS variance estimator is incorrect, even though the OLS coefficient estimator can remain consistent. The asymptotic variance takes the sandwich form

$$\text{Var}(\hat{\beta}_{OLS}) = (X'X)^{-1}X'\Omega X(X'X)^{-1}, \quad (4)$$

up to the corresponding scale factor. This distinction between coefficient estimation and statistical inference motivates the use of generalized least squares and heteroskedasticity-robust variance estimators. The course notes emphasize that non-spherical errors invalidate the conventional variance-covariance matrix of OLS.

2.2 Generalized Least Squares and FGLS

Generalized Least Squares (GLS) provides an alternative to OLS when the covariance structure of the errors is known. Consider

$$E(u \mid X) = 0, \quad E(uu' \mid X) = \sigma^2\Omega. \quad (5)$$

When Ω is known and positive definite, the GLS estimator is

$$\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y. \quad (6)$$

GLS can also be interpreted through a transformation of the original model. If

$$\Omega = PP', \quad (7)$$

then premultiplying the model by P^{-1} gives

$$y^* = X^*\beta + u^*, \quad (8)$$

where

$$y^* = P^{-1}y, \quad X^* = P^{-1}X, \quad u^* = P^{-1}u. \quad (9)$$

The transformed errors have a spherical covariance structure, so GLS is equivalent to applying OLS to the transformed model.

In practice, Ω is generally unknown. When a consistent estimate of Ω is used, the procedure is known as Feasible Generalized Least Squares (FGLS). The course notes define FGLS as estimated GLS and establish its asymptotic equivalence to GLS under the relevant conditions.

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$$\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y. \quad (11)$$

GLS can also be interpreted through a transformation of the original model. If

$$\Omega = PP', \quad (12)$$

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$$y^* = X^*\beta + u^*, \quad (13)$$

where

$$y^* = P^{-1}y, \quad X^* = P^{-1}X, \quad u^* = P^{-1}u. \quad (14)$$

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2.4 White's Test

White's test provides a general test for heteroskedasticity. The null and alternative hypotheses are

$$H_0 : \text{homoskedasticity}, \quad H_1 : \text{heteroskedasticity}. \quad (15)$$

For the model

$$y_i = \alpha_0 + \alpha_1 x_{i1} + \alpha_2 x_{i2} + u_i, \quad (16)$$

the White auxiliary specification relates the conditional error variance to the explanatory variables, their squares, and their cross-product:

$$u_i^2 = \gamma_0 + \gamma_1 x_{i1} + \gamma_2 x_{i2} + \gamma_3 x_{i1}^2 + \gamma_4 x_{i2}^2 + \gamma_5 x_{i1}x_{i2} + v_i. \quad (17)$$

The test statistic is

$$LM = nR^2, \quad (18)$$

where R^2 is obtained from the auxiliary regression. Under the null hypothesis and standard regularity conditions,

$$LM \xrightarrow{d} \chi_q^2, \quad (19)$$

where q is the number of restrictions.

The course notes specify the White test using the explanatory variables, their squares, and their cross-products, with the hypotheses of homoskedasticity versus heteroskedasticity.

2.5 White's Heteroskedasticity-Robust Variance

When heteroskedasticity is present, valid statistical inference can be obtained using a heteroskedasticity-robust estimator of the variance-covariance matrix. The White estimator is

$$\widehat{\text{Var}}(\hat{\beta}_{OLS}) = (X'X)^{-1} \left(\sum_{i=1}^n x_i x_i' \hat{u}_i^2 \right) (X'X)^{-1}, \quad (20)$$

where $\hat{u}_i$ denotes the OLS residual for observation i .

Unlike the conventional variance estimator, this estimator does not require the errors to be homoskedastic. It therefore provides a basis for robust statistical inference when the form of heteroskedasticity is unknown. The course notes identify this estimator as the White (1980) heteroskedasticity-robust variance-covariance matrix.

3 Simulation Design

The finite-sample properties of the econometric procedures considered in this study are evaluated through Monte Carlo simulations. The experiments are designed to vary the sample size and, depending on the exercise, the properties of the error term. For each configuration, 5,000 independent Monte Carlo replications are performed. The simulations use seed 2026 to ensure reproducibility.

The simulation exercises are divided into three components. The first evaluates the finite-sample properties of FGLS and its equivalence with the Cholesky-transformed OLS estimator. The second evaluates the size and power of White's test for heteroskedasticity. The third evaluates the finite-sample accuracy of White's heteroskedasticity-robust variance estimator.

3.1 FGLS Simulation Design

The first experiment considers the cross-sectional model

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad (21)$$

where

$$\beta_0 = -3, \quad x_i \sim U(1, 50). \quad (22)$$

The disturbance is normally distributed with a heteroskedastic variance structure. Five variance levels are considered:

$$\text{Var}(u_i \mid X) \in \{4, 9, 16, 25, 36\}. \quad (23)$$

For a given sample size n , the observations are divided across the five variance groups, with $N = n/5$ observations assigned to each group. Thus, the diagonal elements of the covariance matrix follow the structure

$$\Omega = \text{diag} \left(\underbrace{4, \dots, 4}_N, \underbrace{9, \dots, 9}_N, \underbrace{16, \dots, 16}_N, \underbrace{25, \dots, 25}_N, \underbrace{36, \dots, 36}_N \right). \quad (24)$$

The sample sizes considered are

$$n \in \{5, 10, 30, 100, 200, 500\}. \quad (25)$$

For each replication, the coefficient vector is estimated using the specified covariance structure. The null hypothesis

$$H_0 : \beta_1 = 0.8 \quad (26)$$

is evaluated using the corresponding test statistic. Empirical size is measured under $\beta_1 = 0.8$, while empirical power is evaluated under the alternatives

$$\beta_1 = 0 \quad \text{and} \quad \beta_1 = 0.4. \quad (27)$$

Rejection frequencies are calculated at the 1% and 5% significance levels. The Monte Carlo distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$ are summarized using their mean, median, and standard deviation.

Finally, the same model is transformed using the Cholesky factor of the covariance matrix. OLS is then applied to the transformed model, and the resulting estimates are compared with those obtained from the FGLS procedure.

3.2 White's Test Simulation Design

The second experiment evaluates the finite-sample performance of White's test using the multiple regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \sqrt{\nu_i} \varepsilon_i, \quad (28)$$

with

$$\beta_0 = \beta_1 = \beta_2 = 1. \quad (29)$$

The explanatory variables are constructed from a base sample of 20 observations. The first regressor consists of 20 equally spaced values over the interval $[-1.1, 1.1]$, while the second regressor is generated from a standard normal distribution. These base vectors are replicated to construct the larger sample sizes.

The sample sizes are

$$n \in \{20, 60, 100, 200, 400, 600\}. \quad (30)$$

Three error designs are considered.

Design 0: Homoskedasticity and Normality. Under the null design,

$$\nu_i = 1, \quad \varepsilon_i \sim N(0, 1). \quad (31)$$

Therefore, the disturbances are homoskedastic and normally distributed. This design is used to evaluate the empirical size of White's test.

Design 1: Heteroskedasticity and Normality. The conditional variance is specified as

$$\nu_i = \exp(0.25x_{1i} + 0.25x_{2i}), \quad (32)$$

while the innovations remain normally distributed:

$$\varepsilon_i \sim N(0, 1). \quad (33)$$

The resulting disturbance is

$$u_i = \sqrt{\nu_i} \varepsilon_i. \quad (34)$$

This design is used to evaluate the empirical power of White's test in the presence of heteroskedasticity with normally distributed innovations.

Design 2: Heteroskedasticity and Non-Normality. Design 2 maintains the same heteroskedastic variance structure as Design 1,

$$\nu_i = \exp(0.25x_{1i} + 0.25x_{2i}), \quad (35)$$

but the innovations follow a Student's t distribution with five degrees of freedom:

$$\varepsilon_i \sim t_5. \quad (36)$$

Thus,

$$u_i = \sqrt{\nu_i} \varepsilon_i. \quad (37)$$

This design allows the finite-sample performance of White’s test to be evaluated when heteroskedasticity is combined with non-normal disturbances.

For each replication, White’s test is applied using the auxiliary regression described in Section 2. The empirical rejection rate is calculated at the 1%, 5%, and 10% significance levels. Under Design 0, these rejection rates measure empirical size, whereas under Designs 1 and 2 they measure empirical power.

3.3 White Robust Variance Simulation Design

The third experiment evaluates the finite-sample behavior of White’s heteroskedasticity-robust covariance estimator under Designs 1 and 2. The same regression model, parameter values, explanatory variables, sample sizes, and error structures described above are maintained.

For each Monte Carlo replication, the model is estimated by OLS and the heteroskedasticity-robust covariance matrix is computed using the HC0 estimator. Two versions of the covariance estimator are considered.

First, the standard White estimator is constructed using the squared OLS residuals,

$$\hat{\Omega} = \text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2). \quad (38)$$

Second, the same covariance formula is computed using the squared true simulated disturbances,

$$\Omega^* = \text{diag}(u_1^2, \dots, u_n^2). \quad (39)$$

The resulting variance estimates are compared with the Monte Carlo covariance of the coefficient estimates across the 5,000 replications. The comparison is performed separately for β_0 , β_1 , and β_2 , as well as for the aggregate relative bias across the three coefficient variances.

4 Finite-Sample Properties of FGLS

This section presents the Monte Carlo results for the finite-sample properties of the FGLS estimator. The analysis considers sample sizes $n = 5, 10, 30, 100, 200$, and 500, with 5,000 replications for each configuration.

Three aspects of the estimator are examined. First, the empirical size and power of the hypothesis test for β_1 are evaluated. Second, the sampling properties of the FGLS coefficient estimates are analyzed through their Monte Carlo means, medians, and standard deviations. Finally, the FGLS estimates are compared with those obtained from OLS applied to the Cholesky-transformed model.

4.1 Empirical Size and Power

The first part of the experiment evaluates the finite-sample behavior of the test for the slope coefficient. The null hypothesis is

$$H_0 : \beta_1 = 0.8. \quad (40)$$

Under the null hypothesis, the empirical rejection frequency provides an estimate of the test size. Power is evaluated under the two alternatives

$$\beta_1 = 0 \quad \text{and} \quad \beta_1 = 0.4. \quad (41)$$

Table 1 reports the empirical rejection frequencies at the 1% and 5% significance levels.

Table 1: Empirical Size and Power of the FGLS Test

n	Size 1%	Size 5%	Power 1% ($\beta_1 = 0$)	Power 5% ($\beta_1 = 0$)	Power 1% ($\beta_1 = 0.4$)	Power 5% ($\beta_1 = 0.4$)
5	0.0118	0.0524	0.9410	0.9714	0.6288	0.7766
10	0.0140	0.0514	0.9994	0.9994	0.9474	0.9830
30	0.0110	0.0494	1.0000	1.0000	1.0000	1.0000
100	0.0102	0.0532	1.0000	1.0000	1.0000	1.0000
200	0.0070	0.0430	1.0000	1.0000	1.0000	1.0000
500	0.0110	0.0520	1.0000	1.0000	1.0000	1.0000

The empirical size remains close to the nominal significance levels across the different sample sizes. At the 5% significance level, the rejection rate ranges from 4.30% to 5.32%, while at the 1% level it ranges from 0.70% to 1.40%. The deviations from the nominal levels do not follow a systematic monotonic pattern as the sample size increases.

Figure 1 illustrates these results.

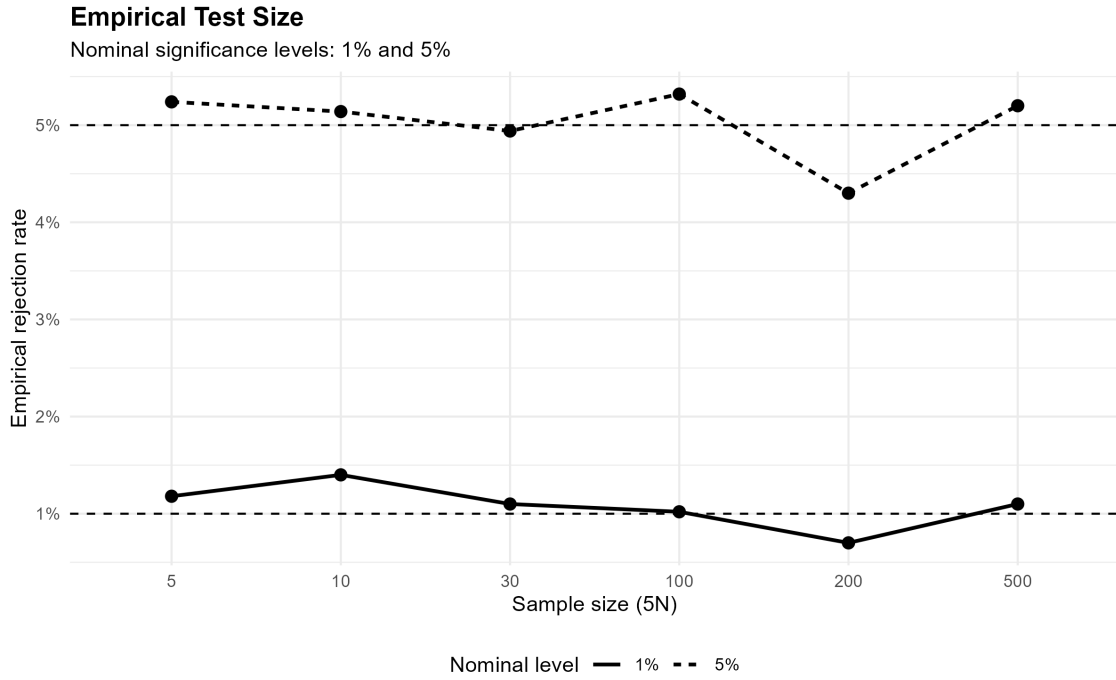


Figure 1: Empirical Test Size of the FGLS Test

The behavior of empirical power is substantially different. When $\beta_1 = 0$, the rejection probability is already high for the smallest sample size, reaching 94.10% at the 1% significance level and 97.14% at the 5% level when $n = 5$. From $n = 30$ onwards, power is equal to 100% at both significance levels.

For the alternative $\beta_1 = 0.4$, power also increases rapidly with the sample size. At the 5% significance level, it rises from 77.66% at $n = 5$ to 98.30% at $n = 10$, reaching 100% from $n = 30$ onwards. At the 1% significance level, the corresponding values are 62.88%, 94.74%, and 100%, respectively.

These results indicate that the test has high ability to detect departures from the null hypothesis under the simulated alternatives, particularly as the sample size increases.

Figure 2 presents the corresponding empirical power curves.

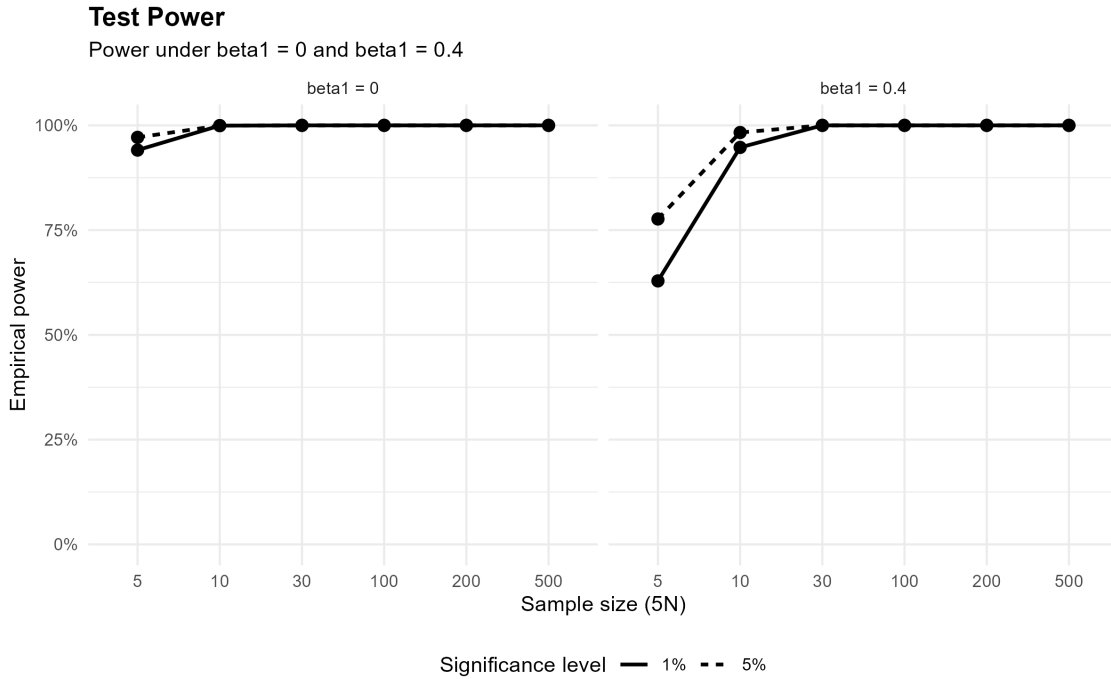


Figure 2: Empirical Power of the FGLS Test

4.2 Sampling Properties of the FGLS Estimator

The second part of the analysis examines the sampling distribution of the FGLS coefficient estimates. Table 2 reports their Monte Carlo means, medians, and standard deviations.

Table 2: Sampling Properties of the FGLS Estimator

n	Mean $\hat{\beta}_0$	Median $\hat{\beta}_0$	SD $\hat{\beta}_0$	Mean $\hat{\beta}_1$	Median $\hat{\beta}_1$	SD $\hat{\beta}_1$
5	-2.9958	-3.0274	4.5188	0.8018	0.8017	0.1626
10	-3.0278	-2.9943	2.4913	0.8013	0.8006	0.0876
30	-2.9966	-2.9656	1.2475	0.7996	0.7997	0.0433
100	-3.0129	-3.0163	0.6758	0.8006	0.8008	0.0232
200	-2.9942	-2.9907	0.4527	0.7997	0.7996	0.0157
500	-3.0047	-3.0043	0.2983	0.8001	0.8000	0.0102

The true parameter values are $\beta_0 = -3$ and $\beta_1 = 0.8$. The Monte Carlo means of both estimators remain close to these values across all sample sizes. For $\hat{\beta}_1$, the estimated mean ranges from 0.7996 to 0.8018, while the mean of $\hat{\beta}_0$ remains close to -3 .

The medians show a similar pattern. In particular, the median of $\hat{\beta}_1$ ranges from 0.7996 to 0.8017, remaining very close to the true value of 0.8 throughout the experiment.

The most pronounced change as the sample size increases is observed in the sampling variability of the estimators. The standard deviation of $\hat{\beta}_1$ decreases from 0.1626 at $n = 5$ to 0.0102 at $n = 500$. Similarly, the standard deviation of $\hat{\beta}_0$ falls from 4.5188 to 0.2983.

Thus, increasing the sample size substantially reduces the dispersion of the FGLS estimates around their corresponding true parameter values.

Figure 3 illustrates the reduction in the sampling variability of $\hat{\beta}_1$.

4.3 FGLS versus Cholesky-Transformed OLS

The final part of the experiment compares the FGLS estimator with OLS applied to the model after the Cholesky transformation. Since the covariance structure of the disturbances is known in the simulation, the model can be transformed using the corresponding Cholesky factor.

The results show that the two approaches produce numerically equivalent coefficient estimates. The comparison is particularly evident for $\hat{\beta}_1$, whose mean and median estimates coincide across the two procedures for all sample sizes considered.

The numerical differences between the two estimators are negligible. The mean absolute differences are approximately of order 10^{-16} , while the maximum differences across replications are of order 10^{-14} , with minor numerical deviations arising only from floating-point computation.

Figure 4 compares the mean and median estimates of $\hat{\beta}_1$ obtained from FGLS and from OLS applied to the Cholesky-transformed model.

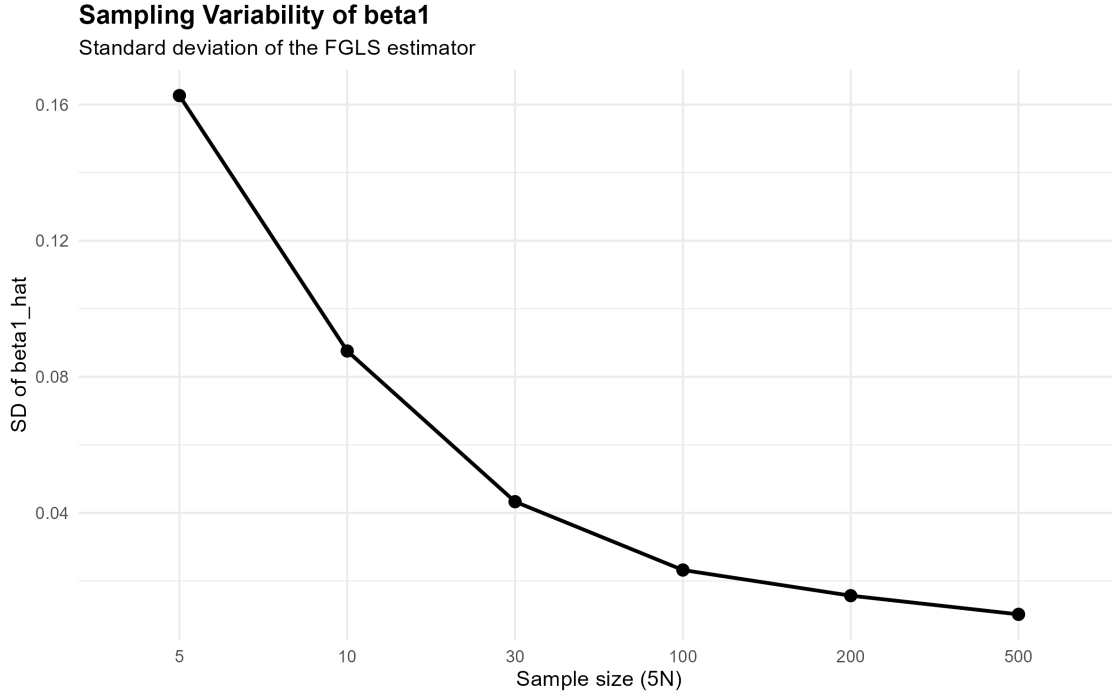


Figure 3: Sampling Variability of $\hat{\beta}_1$

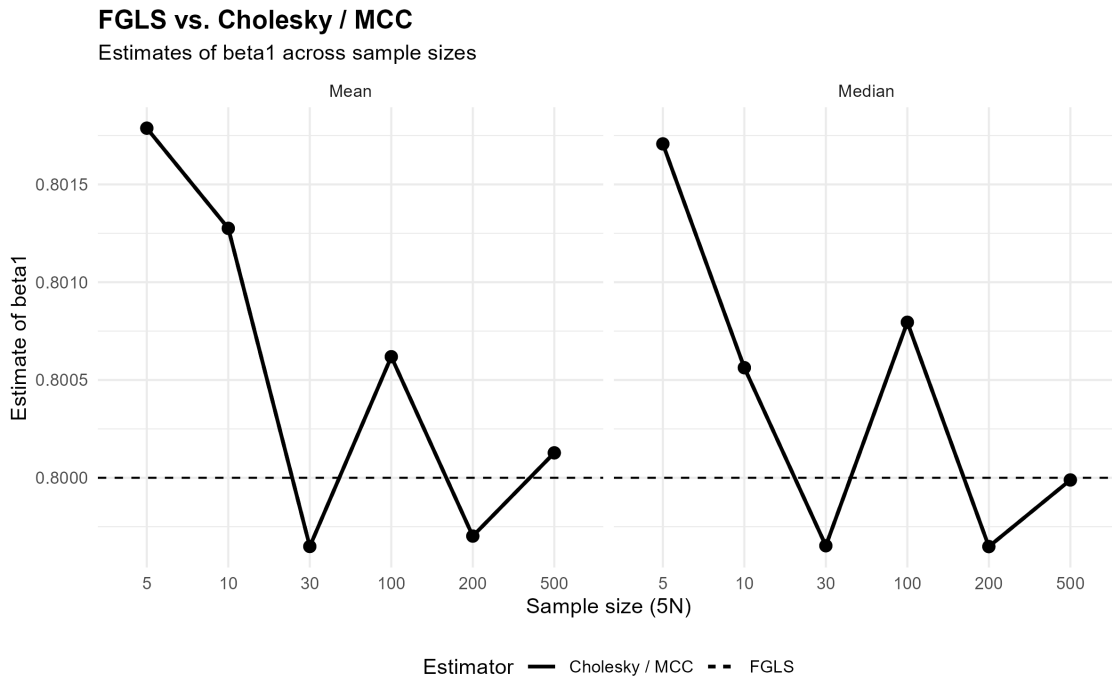


Figure 4: FGLS versus Cholesky-Transformed OLS

The numerical equivalence observed in the simulation is consistent with the algebraic equivalence established in the theoretical framework: GLS can be implemented either directly through the covariance matrix or by

applying OLS to the appropriately transformed model.

5 Finite-Sample Properties of White's Test

This section evaluates the finite-sample performance of White's test for heteroskedasticity. The analysis considers three error designs and sample sizes ranging from $n = 20$ to $n = 600$. For each configuration, 5,000 Monte Carlo replications are performed.

Under Design 0, where the errors are homoskedastic and normally distributed, the empirical rejection frequencies measure the size of the test. Under Designs 1 and 2, which introduce heteroskedasticity, the rejection frequencies measure empirical power. Design 1 combines heteroskedasticity with normally distributed innovations, whereas Design 2 additionally introduces non-normality through Student's t innovations with five degrees of freedom.

5.1 Empirical Size

Under Design 0, the null hypothesis of homoskedasticity is satisfied. Therefore, the empirical rejection frequency should be close to the corresponding nominal significance level.

Table 3 reports the empirical size at the 1%, 5%, and 10% significance levels for all sample sizes.

Table 3: Empirical Size and Power of White's Test

n	Size 1%	Size 5%	Size 10%	Power D1 1%	Power D1 5%	Power D1 10%	Power D2 1%	Power D2 5%	Power D2 10%
20	0.18%	3.86%	8.94%	0.18%	3.98%	9.96%	0.26%	4.60%	10.62%
60	0.82%	4.06%	9.30%	3.30%	11.62%	20.44%	1.48%	7.86%	14.86%
100	0.78%	4.66%	9.98%	5.62%	18.16%	29.32%	2.76%	10.10%	18.08%
200	0.78%	4.74%	9.72%	19.84%	41.74%	54.88%	5.86%	17.98%	28.92%
400	1.16%	4.80%	9.98%	58.56%	81.78%	89.36%	16.82%	36.98%	51.08%
600	1.26%	5.04%	9.72%	87.32%	96.60%	98.38%	30.98%	54.76%	66.96%

The empirical size of White's test is generally close to its nominal level, although the approximation is less accurate for the smallest samples. At the 1% significance level, the empirical rejection rate ranges from 0.18% at $n = 20$ to 1.26% at $n = 600$. At the 5% level, it ranges from 3.86% to 5.04%, while at the 10% level it ranges from 8.94% to 9.98%.

As the sample size increases, the empirical rejection frequencies become closer to their nominal levels. In particular, at $n = 600$, the empirical size is 1.26%, 5.04%, and 9.72% at the 1%, 5%, and 10% levels, respectively.

Figure 5 illustrates the empirical size of White's test under the homoskedastic design.

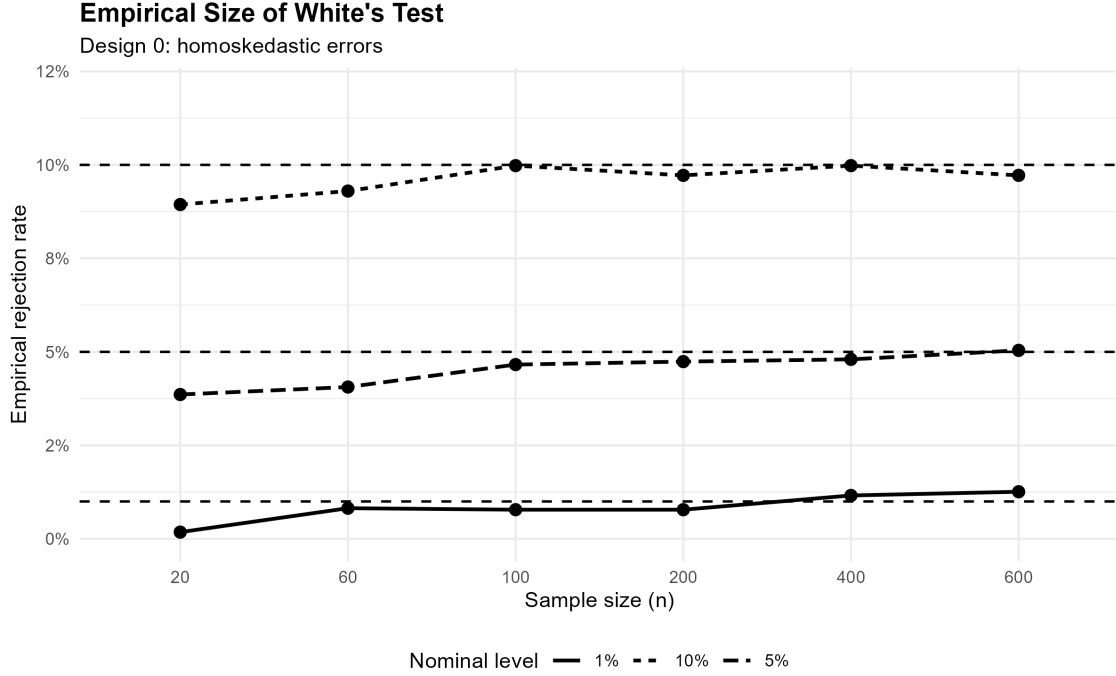


Figure 5: Empirical Size of White's Test

5.2 Empirical Power

The power of White's test is evaluated under Designs 1 and 2. In both cases, the error variance depends on the explanatory variables according to

$$\nu_i = \exp(0.25x_{1i} + 0.25x_{2i}), \quad (42)$$

so that the disturbances are heteroskedastic. The two designs differ in the distribution of the innovations: Design 1 uses normally distributed innovations, whereas Design 2 uses a Student's t distribution with five degrees of freedom.

Under Design 1, empirical power increases substantially with the sample size at all significance levels. At the 5% level, power rises from only 3.98% at $n = 20$ to 11.62% at $n = 60$, 41.74% at $n = 200$, and 96.60% at $n = 600$. The same pattern is observed at the 1% and 10% levels.

The results under Design 2 show a similar increase in power with sample size, but power is systematically lower than under Design 1. At the 5% significance level, for example, power increases from 4.60% at $n = 20$ to 54.76% at $n = 600$. At the 1% level, it rises from 0.26% to 30.98%, while at the 10% level it increases from 10.62% to 66.96%.

The difference between Designs 1 and 2 becomes particularly pronounced for larger samples. At $n = 600$ and the 5% significance level, empirical power reaches 96.60% under Design 1 but only 54.76% under Design 2. Thus, although the presence of heteroskedasticity becomes increasingly detectable as the sample size grows,

the non-normality introduced in Design 2 substantially reduces the finite-sample power of the test.

Figure 6 presents the empirical power curves for both heteroskedastic designs.

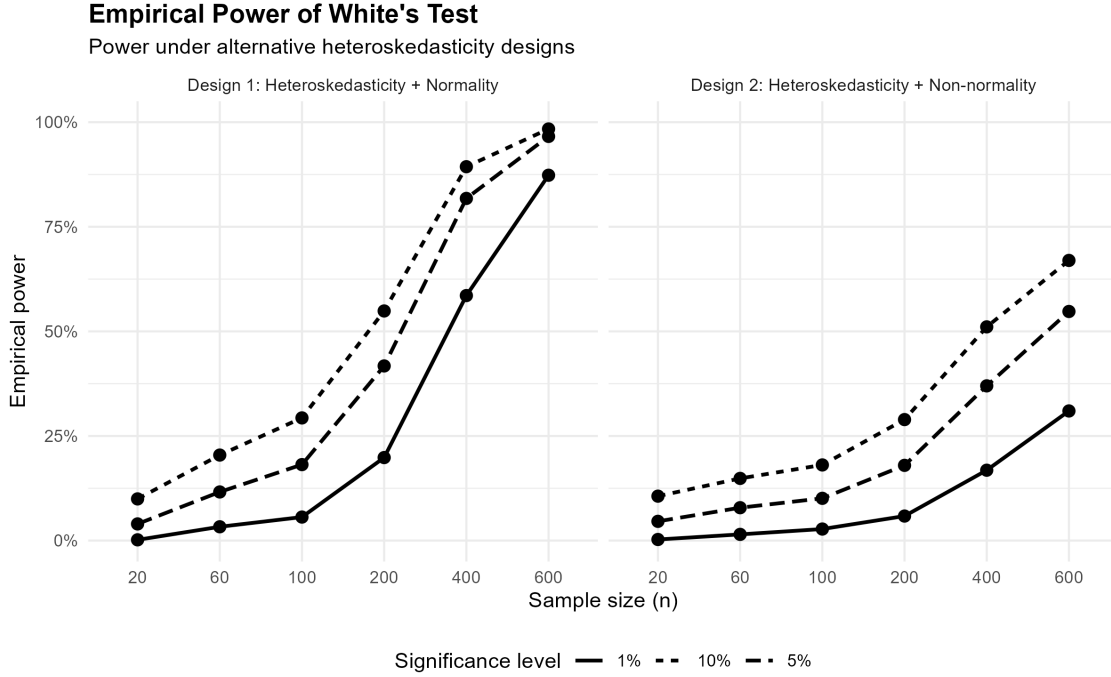


Figure 6: Empirical Power of White's Test

5.3 Finite-Sample Interpretation

The simulation results reveal two distinct aspects of the finite-sample behavior of White's test. First, under the null hypothesis, the empirical size approaches the nominal significance levels as the sample size increases. This indicates that the test exhibits a relatively accurate size in moderate and large samples, while its rejection frequency can deviate more noticeably from the nominal level in very small samples.

Second, the ability of the test to detect heteroskedasticity depends strongly on the sample size and on the distribution of the disturbances. Under Design 1, which combines heteroskedasticity with normal innovations, power increases rapidly with n and becomes very high for the largest samples. Under Design 2, the introduction of non-normal innovations leads to substantially lower power for the same sample sizes.

The comparison between the two designs is particularly informative because the heteroskedastic variance structure is kept unchanged. Therefore, the difference in empirical power reflects the effect of changing the distribution of the innovations from normal to Student's t with five degrees of freedom. The results indicate that this departure from normality can materially affect the finite-sample ability of White's test to detect heteroskedasticity.

6 Finite-Sample Properties of White's Robust Variance

This section examines the finite-sample performance of the heteroskedasticity-robust covariance matrix proposed by White. The analysis focuses on the accuracy of the estimated variances of the OLS coefficients when the disturbances are heteroskedastic. Two alternative data-generating processes are considered, differing in the distribution of the standardized disturbances.

6.1 Simulation Design

The underlying regression model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i, \quad (43)$$

with

$$\beta_0 = \beta_1 = \beta_2 = 1. \quad (44)$$

The sample sizes considered are

$$n \in \{20, 60, 100, 200, 400, 600\},$$

and each configuration is replicated 5,000 times. The regressors are constructed from a fixed base sample of 20 observations and replicated to obtain the different sample sizes. The first regressor is equally spaced over the interval $[-1.1, 1.1]$, while the second regressor is generated from a standard normal distribution.

Two heteroskedastic designs are considered. In both cases, the conditional variance of the disturbance depends on the regressors according to

$$\nu_i = \exp(0.25x_{1i} + 0.25x_{2i}), \quad (45)$$

and the disturbance is generated as

$$u_i = \sqrt{\nu_i} \varepsilon_i. \quad (46)$$

Under Design 1, the standardized disturbance is normally distributed,

$$\varepsilon_i \sim N(0, 1),$$

whereas under Design 2 it follows a Student's t distribution with five degrees of freedom,

$$\varepsilon_i \sim t_5.$$

Thus, Design 1 isolates the effect of heteroskedasticity under normal errors, while Design 2 combines heteroskedasticity with non-normality.

For each simulated sample, the model is estimated by OLS and the heteroskedasticity-robust covariance matrix is computed using White's HC0 estimator. The finite-sample performance of the estimator is evaluated by comparing the average estimated covariance matrix across the Monte Carlo replications with the empirical covariance matrix of the simulated OLS estimates.

For each coefficient, the relative bias of the estimated variance is defined as

$$RB_j = \frac{\overline{\widehat{\text{Var}}(\hat{\beta}_j)} - \text{Var}_{MC}(\hat{\beta}_j)}{\text{Var}_{MC}(\hat{\beta}_j)}, \quad (47)$$

where $\overline{\widehat{\text{Var}}(\hat{\beta}_j)}$ denotes the average White variance estimate across the 5,000 replications and $\text{Var}_{MC}(\hat{\beta}_j)$ denotes the Monte Carlo variance of the corresponding coefficient estimates.

To summarize the overall discrepancy across coefficients, total relative bias is measured as the sum of the absolute relative biases of the three estimated variances.

6.2 Results

The results show that the White covariance estimator exhibits non-negligible finite-sample bias, particularly for the smallest sample sizes. This effect is substantially reduced as the sample size increases.

Figure 7 presents the total relative bias of the variance estimates for the two heteroskedastic designs. Under Design 1, the total bias is particularly large at $n = 20$, where the estimation of the covariance matrix is based on a relatively small number of observations. As the sample size increases, the total bias falls considerably. Although the reduction is not strictly monotonic, the overall pattern indicates that the robust variance estimator becomes more accurate in larger samples.

A similar pattern is observed under Design 2. However, the presence of non-normal disturbances affects the finite-sample behavior of the estimator, especially at smaller and intermediate sample sizes. The total relative bias remains positive throughout the simulations, indicating that, on average, the estimated variances tend to exceed the Monte Carlo variances used as the benchmark.

Figure 8 provides a more detailed view by displaying the relative bias for each individual variance estimate. At $n = 20$, the biases are considerably larger in magnitude, particularly for the intercept variance. As the sample size increases, the individual biases become substantially smaller and generally remain close to zero.

The results also show that the behavior of the individual variance estimates differs across coefficients. The intercept variance tends to display greater finite-sample distortions than the slope variances, which is consistent with the greater sensitivity of the intercept estimate to the finite-sample structure of the regressors. Nevertheless, for the largest sample sizes, the relative biases of all three variance estimates are relatively

small.

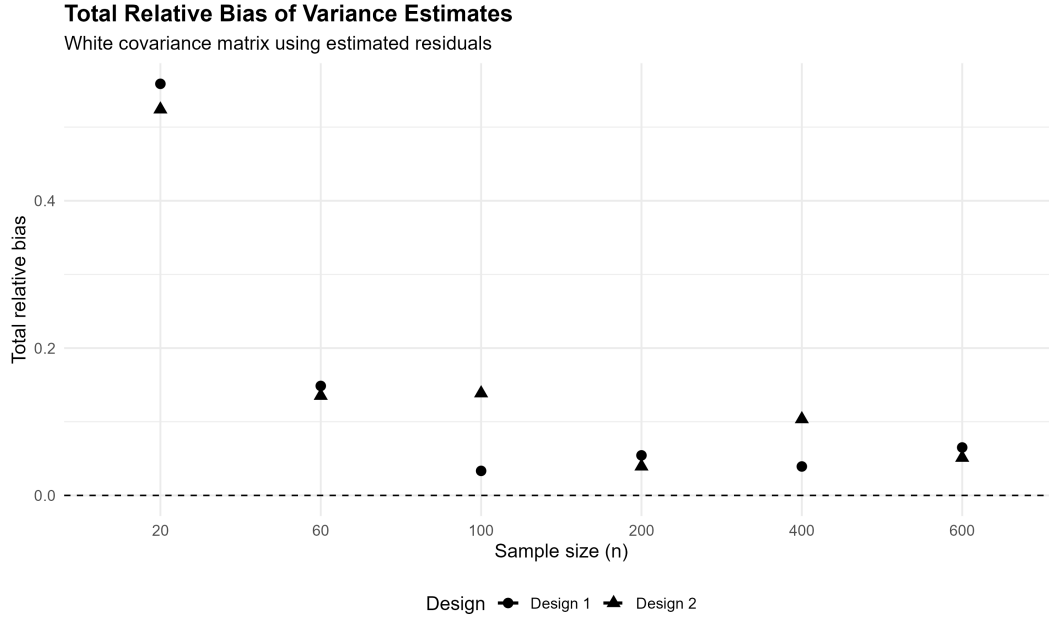


Figure 7: Total relative bias of White variance estimates using estimated residuals.

Note: Total relative bias is the sum of the absolute relative biases of the variance estimates of $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$. Each point is based on 5,000 Monte Carlo replications.

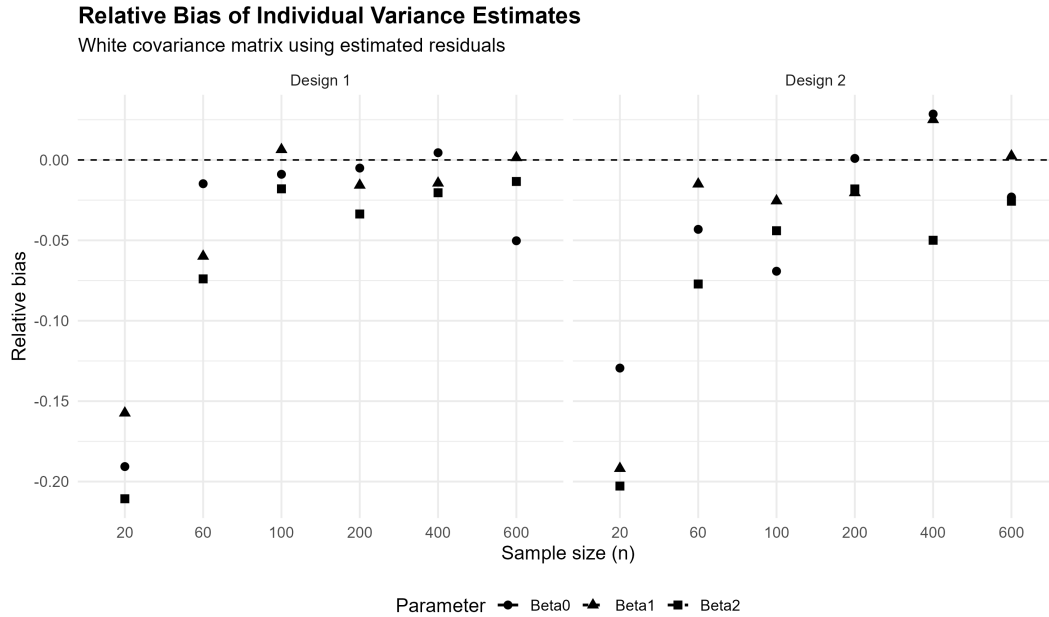


Figure 8: Relative bias of individual White variance estimates using estimated residuals.

Note: The figure reports the relative bias of the variance estimates of $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ under Designs 1 and 2.

Table 4 summarizes the main finite-sample patterns. The results indicate that the accuracy of White's

estimator improves substantially with the sample size. The largest discrepancies occur for $n = 20$, while the relative biases are considerably smaller for $n \geq 100$. This suggests that the asymptotic validity of the heteroskedasticity-robust covariance estimator can be a poor approximation in very small samples, even when the form of heteroskedasticity is correctly handled.

Table 4: Summary of finite-sample behavior of White’s robust variance

Sample size	Design 1	Design 2
Small samples ($n = 20$)	High finite-sample bias	High finite-sample bias
Intermediate samples ($n = 60$ – 200)	Bias decreases substantially	Bias remains somewhat larger
Large samples ($n = 400$ – 600)	Relatively small bias	Relatively small bias

Note: The table summarizes the patterns observed in the Monte Carlo simulations. Exact numerical results are reported in the corresponding output file.

Overall, the simulations indicate that White’s robust covariance estimator performs increasingly well as the sample size grows, but its finite-sample accuracy can be limited in small samples. Importantly, the results should not be interpreted as evidence that the estimator is biased asymptotically. Rather, they illustrate the difference between its finite-sample behavior and the population covariance matrix used as the Monte Carlo benchmark.

7 Estimated Errors versus True Errors

The previous two sections evaluated the finite-sample properties of White’s heteroskedasticity-robust covariance estimator under two different implementations. Section 6 considered the standard empirical case, in which the unobserved disturbances are replaced by estimated OLS residuals. In this section, the resulting variance estimates are compared with those obtained when the true disturbances generated by the data-generating process are used.

The purpose of this comparison is to quantify the finite-sample effect of replacing the unobserved errors with estimated residuals. Since the true disturbances are available in the Monte Carlo experiment, they provide a benchmark against which the additional discrepancy introduced by residual estimation can be evaluated.

7.1 Comparison of Total Relative Bias

For each design and sample size, the total relative bias obtained using estimated residuals is compared with the corresponding value obtained using the true disturbances. Total relative bias is defined as the sum of the absolute relative biases of the estimated variances of the three OLS coefficients,

$$RB_{\text{total}} = |RB_0| + |RB_1| + |RB_2|. \quad (48)$$

The comparison is reported in Table 5. The results indicate that the use of estimated residuals can generate substantially larger finite-sample discrepancies, particularly for small sample sizes.

The contrast is particularly pronounced at the smallest sample size. Under Design 1, total relative bias decreases from 0.5587 when estimated residuals are used to 0.0684 when true errors are used, representing a

Table 5: Total relative bias: estimated residuals versus true errors

n	Design	Estimated residuals	True errors	Difference
20	1	0.5587	0.0684	0.4904
20	2	0.5241	0.0297	0.4944
60	1	0.1486	0.0703	0.0783
60	2	0.1353	0.0293	0.1060
100	1	0.0333	0.0648	-0.0315
100	2	0.1387	0.0671	0.0716
200	1	0.0544	0.0590	-0.0046
200	2	0.0394	0.0416	-0.0023
400	1	0.0393	0.0357	0.0036
400	2	0.1035	0.0589	0.0447
600	1	0.0651	0.0753	-0.0102
600	2	0.0512	0.0173	0.0339

Note: Difference is calculated as total relative bias using estimated residuals minus total relative bias using true errors. Each value is based on 5,000 Monte Carlo replications.

reduction of approximately 87.8%. Under Design 2, the corresponding values are 0.5241 and 0.0297, implying a reduction of approximately 94.3%.

The difference remains substantial at $n = 60$. Under Design 1, total relative bias decreases from 0.1486 to 0.0703, corresponding to a reduction of approximately 52.7%. *Under Design 2, it decreases from 0.1353 to 0.0293, representing a*

For larger samples, however, the relative ranking of the two approaches is not uniform. Under Design 1, the total relative bias obtained using estimated residuals is lower than that obtained using true errors at $n = 100$ and $n = 600$. At $n = 200$ and $n = 400$, the two values are also very close. Under Design 2, the true-error specification produces lower total relative bias at $n = 100$, $n = 400$, and $n = 600$, while the difference is slightly reversed at $n = 200$.

Figure 9 illustrates the total relative bias obtained when the true disturbances are used across sample sizes and error designs. The figure shows that the magnitude of the bias remains relatively limited, although it fluctuates across sample sizes rather than following a strictly monotonic pattern. Under Design 1, the total relative bias ranges from 0.0357 to 0.0753 for the larger sample sizes, while under Design 2 it ranges from 0.0173 to 0.0671. Overall, the results suggest that when the true disturbances are used, the contribution associated with estimating the residuals is removed, leading to substantially smaller discrepancies in the smallest samples.

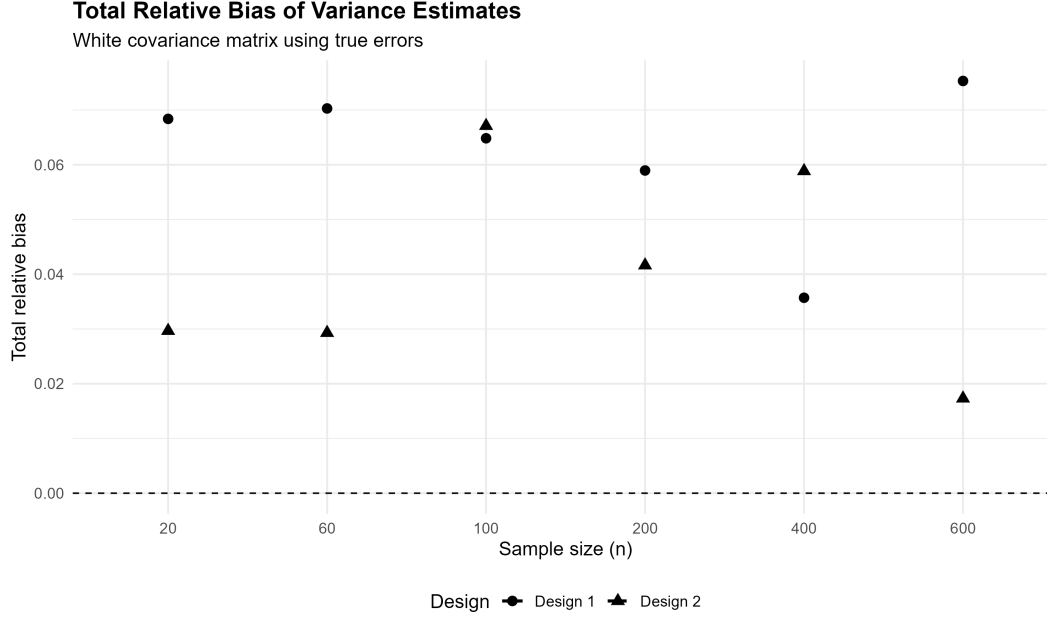


Figure 9: Total relative bias of White variance estimates using true errors.

Note: Total relative bias is defined as the sum of the absolute relative biases of the variance estimates of $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$. Each point is based on 5,000 Monte Carlo replications.

The comparison therefore indicates that the effect of replacing the true disturbances with estimated residuals is most pronounced in small samples. As the sample size increases, the difference between the two approaches becomes considerably smaller and varies across designs and sample sizes. Thus, the results do not support a uniformly lower bias when true errors are used for every sample size, but they clearly show the importance of residual estimation for the finite-sample performance of White's robust variance estimator.

7.2 Individual Variance Estimates

The comparison can also be examined at the level of the individual coefficient variance estimates. Figures 8 and 10 report the relative bias of the variance estimates for $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ using estimated residuals and true errors, respectively.

Figure 10 shows that, when the true disturbances are used, the individual relative biases are generally closer to zero than those observed with estimated residuals in the smallest samples. The difference is particularly pronounced at $n = 20$, where all three variance estimates obtained with estimated residuals display negative relative bias of considerable magnitude.

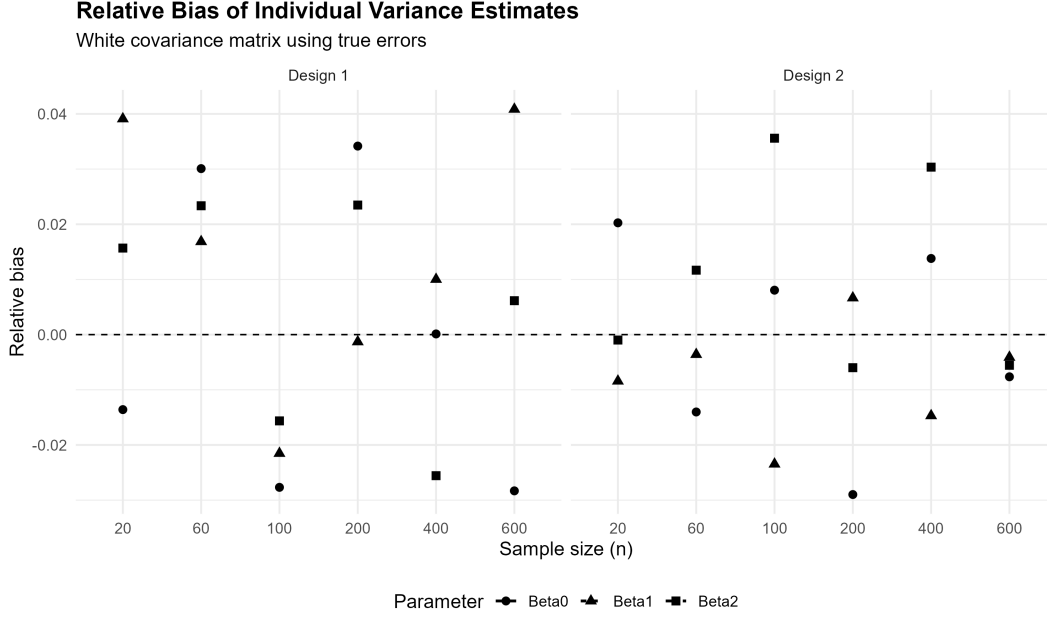


Figure 10: Relative bias of individual White variance estimates using true errors.

Note: The figure reports the relative bias of the variance estimates of $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ under Designs 1 and 2. Each point is based on 5,000 Monte Carlo replications.

When estimated residuals are used, the corresponding figure reported in Section 6 shows substantially larger deviations from zero at $n = 20$. For Design 1, the relative biases of the variance estimates of $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ are -0.1907 , -0.1574 , and -0.2107 , respectively. Under Design 2, the corresponding values are -0.1294 , -0.1919 , and -0.2028 .

In contrast, when true errors are used, the relative biases at $n = 20$ are much smaller. Under Design 1, they are -0.0136 , 0.0391 , and 0.0157 , respectively. Under Design 2, they are 0.0203 , -0.0084 , and -0.0010 .

This comparison indicates that the main source of the large discrepancies observed in the smallest samples is not the heteroskedastic structure itself, but the replacement of the unobserved disturbances by estimated residuals.

7.3 Evolution with Sample Size

The difference between the two approaches becomes less pronounced as the sample size increases, although the pattern is not strictly monotonic. With estimated residuals, total relative bias falls sharply between $n = 20$ and $n = 60$ under both designs. Under Design 1, it declines from 0.5587 to 0.1486 , while under Design 2 it falls from 0.5241 to 0.1353 .

For larger samples, both approaches generally produce variance estimates that are relatively close to the Monte Carlo benchmark. However, the observed values fluctuate across sample sizes. For example, under Design 2, the total relative bias using estimated residuals is 0.1387 at $n = 100$, decreases to 0.0394 at $n = 200$, increases to 0.1035 at $n = 400$, and then falls to 0.0512 at $n = 600$.

A similar pattern is observed when true errors are used. Under Design 1, total relative bias ranges between

0.0357 and 0.0753 for the largest three sample sizes. Under Design 2, it ranges between 0.0173 and 0.0589 over the same sample sizes.

Therefore, the Monte Carlo results should not be interpreted as evidence of a strictly monotonic convergence of the relative bias to zero. The simulation contains a finite number of replications, and the estimated relative biases naturally fluctuate across samples. The important result is the substantially larger discrepancy generated by estimated residuals in small samples.

7.4 Role of the Error Distribution

The comparison between Designs 1 and 2 also provides information about the role of the error distribution. Design 1 combines heteroskedasticity with normally distributed disturbances, whereas Design 2 combines the same heteroskedastic variance structure with non-normal Student's t disturbances.

When estimated residuals are used, Design 2 generally exhibits a substantial total relative bias, particularly at $n = 20, 100$, and 400 . However, the non-normality does not produce a uniformly larger bias across all sample sizes.

The same conclusion applies when true errors are used. For example, total relative bias under Design 2 is smaller than under Design 1 at $n = 20, 60$, and 600 , but larger at $n = 100, 200$, and 400 . Thus, the simulation does not provide evidence of a systematic ranking between the two designs in terms of variance-estimation bias.

Instead, the results suggest that the finite-sample consequences of using estimated residuals are strongly dependent on sample size and, to a lesser extent, on the underlying distribution of the disturbances.

7.5 Overall Comparison

The comparison between estimated residuals and true errors leads to three main findings.

First, the use of estimated residuals generates substantially larger relative bias in small samples. At $n = 20$, total relative bias is more than eight times larger under Design 1 and more than seventeen times larger under Design 2 when estimated residuals are used instead of true errors.

Second, the difference between the two approaches becomes considerably smaller in larger samples. Although the relative bias does not decrease monotonically in every simulation point, the very large discrepancies observed at $n = 20$ are no longer present at the largest sample sizes.

Third, the results emphasize the distinction between a theoretical benchmark and an empirical estimator. The true-error covariance matrix cannot be implemented in an empirical application because the disturbances are unobserved. Nevertheless, its inclusion in the Monte Carlo experiment makes it possible to isolate the finite-sample effect of estimating the disturbances.

Overall, the evidence indicates that replacing the true disturbances with estimated residuals can generate a substantial finite-sample distortion in heteroskedasticity-robust variance estimates, especially when the sample size is small. As the sample size increases, this additional discrepancy becomes considerably less pronounced, although the Monte Carlo results do not imply a strictly monotonic convergence pattern.

8 Comparative Discussion

The Monte Carlo experiments provide complementary evidence on the finite-sample performance of estimation, specification testing, and robust inference under heteroskedasticity. The results from the three parts of the study highlight different consequences of the same underlying problem: the efficiency of estimation when the covariance structure is known, the ability to detect heteroskedasticity when its form is unknown, and the accuracy of variance estimation when robust inference is required.

The results for FGLS show that, when the variance structure of the disturbances is correctly specified, the estimator performs well across the considered sample sizes. The Monte Carlo means of the estimated coefficients remain close to their true values, while their sampling variability decreases substantially as the sample size increases. The empirical size of the test remains close to the nominal significance levels, although some deviations are observed in the smallest samples. At the same time, statistical power increases rapidly with sample size. These results are consistent with the expected behavior of FGLS when the covariance matrix of the disturbances is correctly specified.

The comparison between FGLS and OLS applied to the Cholesky-transformed model provides an additional check of the implementation. The two procedures produce virtually identical coefficient estimates, with differences attributable only to numerical precision. This confirms the theoretical equivalence between GLS and OLS after applying the appropriate transformation to the model. Thus, the first simulation experiment illustrates that knowledge of the heteroskedastic structure can be exploited to obtain efficient estimation and reliable inference.

The results for White's test illustrate a different aspect of the problem. Under the homoskedastic design, the empirical rejection frequencies remain reasonably close to their nominal levels, especially for the larger samples. Under the heteroskedastic designs, however, the ability of the test to detect the violation of homoskedasticity depends strongly on the sample size. Power is limited in the smallest samples and increases substantially as the number of observations grows. This shows that the absence of rejection in a small sample should not necessarily be interpreted as evidence that the disturbances are homoskedastic.

The comparison between the two heteroskedastic designs is particularly informative. White's test has substantially greater power under Design 1, where the disturbances are normally distributed, than under Design 2, where the disturbances follow a t_5 distribution. At the 5% significance level, power under Design 1 increases from 3.98% at $n = 20$ to 96.60% at $n = 600$, whereas under Design 2 it increases from 4.60% to 54.76%. Therefore, the non-normality of the disturbances has an important effect on the finite-sample performance of the test. The results emphasize that the usefulness of a specification test depends not only on the presence of the feature being tested, but also on the sample size and the distributional characteristics of the disturbances.

The results for White's robust variance estimator provide a further perspective on the finite-sample consequences of heteroskedasticity. When estimated residuals are used, the total relative bias is particularly large in the smallest samples. At $n = 20$, the total relative bias is 0.5587 under Design 1 and 0.5241 under Design 2. The magnitude of the bias becomes considerably smaller at larger sample sizes, although the evolution is not strictly monotonic. This indicates that residual estimation can have an important effect on the finite-sample accuracy of robust variance estimates, particularly when the number of observations is limited.

The comparison with the benchmark based on the true disturbances provides additional evidence on this issue.

At $n = 20$, replacing estimated residuals with the true disturbances reduces total relative bias from 0.5587 to 0.0684 under Design 1 and from 0.5241 to 0.0297 under Design 2. These differences become considerably smaller as the sample size increases. However, the ranking of the two approaches is not uniform across all sample sizes and designs. In some cases, the estimator based on estimated residuals produces a lower total relative bias than the benchmark based on true disturbances. Therefore, the true-error experiment should be interpreted as an artificial benchmark that isolates the contribution of residual estimation, rather than as evidence that true disturbances uniformly improve finite-sample performance.

Taken together, the three experiments reveal an important distinction between estimation, diagnostic testing, and robust inference. FGLS can perform very well when the covariance structure is correctly known, but this advantage relies on information about the form of heteroskedasticity. White’s test provides a general diagnostic for heteroskedasticity, but its finite-sample power can be limited, especially under non-normal disturbances. White’s robust variance estimator does not require the covariance structure to be explicitly specified, making it useful when the form of heteroskedasticity is unknown, but its finite-sample accuracy can be affected by the use of estimated residuals.

The combined evidence therefore suggests that the choice of procedure should depend on the information available about the disturbance process and on the characteristics of the sample. When the variance structure is known and correctly specified, FGLS offers an efficient approach. When the structure is unknown, specification tests such as White’s test can provide useful diagnostic information, although their power should be interpreted with caution in small samples. When robust inference is required, White’s variance estimator provides a practical alternative, but its finite-sample properties should be distinguished from its asymptotic justification.

Overall, the simulations show that sample size plays a central role in the performance of all three procedures. The results also demonstrate that the distribution of the disturbances matters for statistical testing, while the estimation of the disturbances themselves can be particularly relevant for robust variance estimation. These findings highlight why the theoretical properties of econometric procedures should be complemented by an assessment of their finite-sample behavior.

9 Conclusion

This study examined the finite-sample properties of FGLS, White’s heteroskedasticity test, and White’s heteroskedasticity-robust variance estimator through a set of Monte Carlo experiments. The simulations were designed to evaluate how sample size, heteroskedasticity, and the distribution of the disturbances affect estimation and inference.

The results show that FGLS performs well when the covariance structure of the disturbances is correctly specified. The estimated coefficients remain close to their true values, sampling variability decreases with sample size, and the empirical size and power of the associated tests behave as expected. The numerical equivalence between FGLS and OLS applied to the Cholesky-transformed model also confirms the consistency between the theoretical formulation and its computational implementation.

White’s test exhibits adequate empirical size under homoskedasticity, but its power depends strongly on sample size and on the distribution of the disturbances. In particular, the test is substantially more powerful

under normally distributed disturbances than under the t_5 specification considered in the simulations. This result illustrates the importance of finite-sample considerations when interpreting the outcome of specification tests.

The simulations also show that White's robust variance estimator can exhibit considerable finite-sample bias when estimated residuals are used, particularly in small samples. The comparison with the true-disturbance benchmark indicates that residual estimation accounts for an important part of this discrepancy in the smallest samples, although the difference becomes less systematic as the sample size increases.

Overall, the evidence highlights that the choice between FGLS, diagnostic testing, and robust inference cannot be based exclusively on their theoretical or asymptotic properties. Sample size, the distribution of the disturbances, and the available information about the variance structure all influence their finite-sample performance. The Monte Carlo approach therefore provides a useful complement to asymptotic econometric theory by showing how these procedures behave in settings that more closely resemble the limitations encountered in finite samples.